SUMMER INTERNSHIP REPORT

GenAI Powered Data Analytics – Job Simulation

Organization: Tata iQ (via Forage)

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Session	2025–29
Semester	3rd Semester
Section	2CSE12
Completion	September 20, 2026

Note: This report documents the Tata – GenAI Powered Data Analytics Job Simulation completed through Forage. It describes the simulation work and learning outcomes and does not represent employment at Tata.

1. Introduction

The GenAI Powered Data Analytics Job Simulation provided practical exposure to AI-powered data analytics and strategy development for a Financial Services team at Tata iQ. The simulation focused on using data and GenAI-assisted reasoning to understand customer delinquency risk, structure a predictive modelling approach, and design an AI-driven collections strategy.

2. Objectives

The main objectives were to conduct exploratory data analysis (EDA), assess data quality and identify potential risk indicators, propose an initial predictive modelling framework for customer delinquency, translate analytical findings into business recommendations, and design a scalable AI-powered collections approach with responsible-AI safeguards.

3. Exploratory Data Analysis and Data Quality

The provided Geldium customer dataset contained 500 records and 19 columns. The target variable, Delinquent_Account, indicated a 16% delinquency rate (80 of 500 records). No duplicate rows or duplicate Customer_ID values were identified. Missing values were present in Income (39 records), Credit_Score (2 records), and Loan_Balance (29 records). The four Credit_Utilization values above 1.0 should be validated because utilization is generally represented as a ratio or percentage.

3.1 Key Observations

Employment_Status contained inconsistent labels such as EMP, employed and Employed; these should be normalized before modelling. Descriptive analysis considered recent payment behaviour (Month_1–Month_6 and Missed_Payments), Credit_Utilization, Debt_to_Income_Ratio, Credit_Score, Income and Loan_Balance as candidate risk indicators. Simple Pearson correlations with delinquency were weak, so these variables should be treated as candidates for formal model validation rather than proven causal predictors.

3.2 Additional Data Insight

Customers with missing Loan_Balance had an observed delinquency rate of approximately 24.1%, compared with

approximately 15.5% among customers whose Loan_Balance was present. This difference suggests that missingness itself may carry information and should be investigated rather than handled blindly. The dataset also has a minority positive class, making class-aware evaluation important.

4. Predictive Modelling Framework

A Logistic Regression model was selected as the initial model because the target is binary and the model can produce an interpretable delinquency probability. A practical workflow is: validate and clean the data; impute missing numeric values; normalize inconsistent categorical labels; encode categorical variables; standardize appropriate numeric features; use a stratified train/test split and cross-validation; train the model; and evaluate predicted probabilities.

4.1 Candidate Features

Candidate features include Missed_Payments, Month_1–Month_6 payment history, Credit_Utilization, Debt_to_Income_Ratio, Credit_Score, Income and Loan_Balance. Customer_ID should be excluded because it is an identifier rather than a behavioural risk feature. Age, Account_Tenure, Location and Credit_Card_Type may be considered as secondary variables subject to relevance, proxy-risk and fairness review.

4.2 Model Evaluation

Because only 16% of observations are delinquent, accuracy alone is not sufficient. Evaluation should include precision, recall, F1 score, AUC-ROC, confusion matrix and probability calibration. Threshold selection should reflect the cost of missed high-risk customers, unnecessary outreach and available collections capacity. A Decision Tree can serve as a challenger model; more complex models should only be introduced if they provide material validated improvement while remaining explainable and governable.

5. Business Recommendation

The proposed intervention is an early-outreach pilot for customers assessed as higher risk. The objective is to contact customers before delinquency becomes more severe using supportive reminders, repayment guidance and appropriate repayment arrangements. A SMART pilot target is a 10% relative reduction in 30+ day delinquency compared with a comparable control group over 12 weeks, with reviews at weeks 4, 8 and 12.

6. AI-Powered Collections Strategy

The proposed system combines customer data, predictive risk scoring and business rules to select appropriate actions. The workflow is: data input → risk scoring and decision logic → intervention/action → outcome monitoring and learning. Lower-risk cases can receive automated reminders and general repayment guidance, while sensitive or complex cases should be routed to human staff.

6.1 Agentic AI and Human Oversight

Agentic AI can support automated, context-aware follow-ups, reminders, general offers and low-risk collections activities. Human oversight should remain in place for tailored debt restructuring, legal escalation, high-risk cases, complaints and other sensitive decisions. The system should record actions and outcomes so performance can be monitored and the intervention strategy improved responsibly.

7. Responsible AI

Key safeguards include fairness monitoring across relevant customer groups, explainable risk flags, data-quality checks, privacy protection, appropriate access controls and compliance with applicable consumer and data-protection requirements. Performance should be compared across groups using measures such as precision, recall and false-positive/false-negative rates, with disparate-impact-style checks where appropriate. Automated outputs should support—not replace—human decision-making in critical cases.

8. Skills Developed

The simulation strengthened skills in AI analytics, AI strategy, analytical reporting, business communication, data interpretation, data quality management, decision making, ethical reasoning, exploratory data analysis, model selection, model validation, predictive analytics, process automation, regulatory compliance and strategic thinking.

9. Conclusion

The Tata – GenAI Powered Data Analytics Job Simulation demonstrated how analytics, predictive modelling and GenAI-assisted strategy can be connected to a practical financial-services use case. The work emphasized not only prediction, but also data quality, interpretability, fairness, responsible automation and business impact. The experience provided a structured foundation for applying data analytics and AI concepts to real-world decision-support problems.